

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MMLU | Context: Non-Arab Tourists and Expats in Eight Arab Countries

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU's single-answer four-option MCQ schema [Q49, Q56] is structurally incompatible with the deployment's required output behavior — acknowledging multiple perspectives on contested topics (1948 war/Nakba, Western Sahara, Lebanese Civil War, Palestinian statehood, Jerusalem status), per elicitation Q3 and the contested-topics YAML section. Every sampled MMLU example has exactly one labeled correct answer with no mechanism for multi-perspective or uncertainty-flagging responses. The authors note near-random model accuracy on 'morality and law' [Q8, Q14] but do not address the structural taxonomy issue. No identified Arab-focused benchmark uses multi-perspective scoring (FV5). Given OO is HIGH priority, this is a fundamental structural-validity violation.

ID	Evidence
Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q8	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"Our results indicate that while recent advances have been impressive, state-of-the-art models still struggle at learning and applying knowledge from pretraining. The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)

Strengths

- Difficulty stratification (Elementary/High School/College/Professional) [Q26] does provide useful granularity within the MCQ schema
- Aggregation across four broad clusters (humanities, social sciences, STEM, other) [Q99–Q101] supports diagnostic interpretation

ID	Evidence
Q26	"Some tasks cover a subject, like psychology, but at a specific level of difficulty, such as "Elementary," "High School," "College," or "Professional."" (p.3)
Q99	"RoBERTa-base attains an overall accuracy of 27.9%, with 27.9% accuracy for the humanities, 28.8% for social sciences, 27.0% for STEM, and 27.7% for other." (p.12)
Q100	"ALBERT-xxlarge attains an accuracy of 27.1%, with 27.2% accuracy for the humanities, 25.7% for the social sciences, 27.7% for STEM, and 27.9% for other." (p.12)
Q101	"GPT-2 attains an accuracy of 32.4%, with 32.8% accuracy for the humanities, 33.3% for the social sciences, 30.2% for STEM, and 33.1% for other." (p.12)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: MCQ A/B/C/D label categories are regionally neutral in form but cannot encode multi-perspective or contested-topic acknowledgment, which the deployment explicitly requires per elicitation Q3.

OO-2: Missing categories: no schema slot for 'this is contested,' 'depends on country,' or 'multiple perspectives apply' — required for Palestinian narrative, Western Sahara, Lebanese Civil War, etc. [contested_topics YAML section].

OO-3: moral_scenarios schema explicitly encodes 'US standards as of 2020' as the decision rule (D12), a non-regional value framing.

OO-4: Stakeholder-driven redesign would require moving beyond MCQ entirely; no Arab-focused benchmark using multi-label or open-ended contested-history scoring has been identified (FV5).

OO-5: Documented structural-validity violation: the construct (multi-perspective handling of contested Arab regional topics) cannot be represented in the MMLU output schema; MMLU paper itself flags weak performance on 'human values such as law and morality' [Q14, Q15] without addressing the schema issue.

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Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
Q8	"Worse, they still have near-random accuracy on some socially important subjects such as morality and law." (p.1)
Q14	"Our results indicate that while recent advances have been impressive, state-of-the-art models still struggle at learning and applying knowledge from pretraining. The tasks with near-random accuracy include calculation-heavy subjects such as physics and mathematics and subjects related to human values such as law and morality." (p.2)
Q15	"This second weakness is particularly concerning because it will be important for future models to have a strong understanding of what is legal and what is ethical." (p.2)
Q49	"To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks." (p.5)
Q56	"We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with 'The following are multiple choice questions (with answers) about [subject].' For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with 'Answer: '. The model then produces probabilities for the tokens 'A,' 'B,' 'C,' and 'D,' and we treat the highest probability option as the prediction. For consistent evaluation, we create a dev set with 5 fixed few-shot examples for each subject." (p.6)
D12	For which of these two scenarios does the main character do something clearly morally wrong, according to ordinary moral standards in the US as of 2020? — Explicitly frames moral judgment via US 2020 standards
WEB-5	ILMAAM (2025) identified significant cultural misalignments in religion and morality content; 11 expert annotators reviewed for cultural appropriateness, bias, religious sensitivity (https://aclanthology.org/2025.loreslm-1.29/)
WEB-11	MENAValues evaluates LLMs across neutral, personalized, and observer framings — illustrating richer output-form evaluation that MMLU's MCQ accuracy cannot match (https://arxiv.org/html/2510.13154)

Information Gaps

- Whether any planned MMLU successor or supplement supports multi-perspective scoring (none identified per FV5)

Requires Expert Verification

- Region-stakeholder design of an output schema that accommodates contested-topic acknowledgment

Remediation

Gap 1: Single-answer MCQ schema cannot represent multi-perspective handling of contested topics (1948 war, Western Sahara, Lebanese Civil War, Palestinian statehood, Jerusalem) — the deployment's explicitly required output behavior per elicitation Q3.

Recommendation: Develop a custom evaluation harness for contested topics using multi-perspective rubrics with Arab regional expert adjudication; do not rely on MMLU MCQ accuracy for these topics. MENAValues [\[WEB-11\]](#) cross-lingual framing methodology is the closest published model.

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Q3	"We design the benchmark to measure knowledge acquired during pretraining by evaluating models exclusively in zero-shot and few-shot settings." (p.1)
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